Baptiste AUDUGÉ

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Engineering Student at CentraleSupélec | Machine Learning & Quantitative Modelling

Seeking a 6-month internship in London, from February 2027

Engineering student in maths & CS who builds tools people rely on: deep learning models on 20 years of uranium mining data at Orano, financial tools used by brokers and advisors in client meetings, and statistical models of dependencies across MSCI World stocks.

EXPERIENCE

Orano | *Global nuclear fuel cycle company* — Paris, France
Data Scientist Intern – Deep Learning — Jul 2026 – Jan 2027

- Asked to fix a **production AI model** planning well cleaning across uranium mines; working with mine engineers and the data team, traced its errors to an overlooked factor: **interactions between neighbouring wells**.
- Built the **data extraction pipeline** (SQL) from the mine's monitoring systems; found **~30% of legacy data silently imputed**.
- Designed a **physics-informed model** capturing well-to-well interactions, grounded in the geology literature.
- Implementing the model in PyTorch (Linux, Git): Transformer with spatio-temporal cross-attention coupled to a Neural ODE, trained on an NVIDIA DGX Spark on **10,000+ well time series** spanning up to **20 years**.

QSTNMRK | *Art Collective (Freelance)* — Paris, France
Lead Developer — Jan 2024 – Present

- Promoted from website rebuild to lead developer; built The Last Dollar (2026, featured in designboom), a **24/7 live-streamed sculpture**, end to end, including the streaming pipeline (MediaMTX, HLS) kept online at low hosting cost.

Freelance Full-Stack Developer — Paris, France
Monaliza | *B2B2C financial products distributor, in partnership with* **Generali** — Jun 2022 – Sep 2024

- Built with the founders the projection tool (NestJS, React) **brokers used in client meetings** to present the company's retirement products; also modelled the **B2B commission schemes** (upfront, trail and linear payouts, clawbacks).

Lexem | *Wealth Management Firm* — Nov 2020 – Jul 2022

- Built with advisors a r**etirement-plan (PER) simulator** projecting capital at retirement and each client's **tax saving**, including carried-over deduction ceilings.

EDUCATION

CentraleSupélec – Université Paris-Saclay | *MSc-level Engineering Degree* — Paris, France
Coursework: Machine Learning, High-Performance Computing, Statistics, Parallel Algorithms — 2024 – 2028 (expected)

Sorbonne University | *Double Bachelor's Degree in Mathematics & Computer Science* — Paris, France
Coursework: Advanced C Programming (ranked 1st/361), Data Structures, Databases, Probability, Algebra — 2021 – 2024

ACADEMIC PROJECTS

- **Graphical Models in Finance:** sparse dependency networks across **MSCI World stocks** (Graphical Lasso, Ledoit-Wolf) on rolling 250-day windows; networks densify with new cross-sector links in turbulent periods.
- **HPC Aircraft Drag Simulation:** CFD simulation parallelised with **MPI/OpenMP** and run on a **SLURM**-managed cluster.
- **Picross Solver (C++):** nonogram solver combining line-by-line **dynamic programming** for constraint propagation with backtracking and forward checking when propagation stalls; built with CMake.
- **MyGit (C):** reimplemented a simplified Git in C with **no external libraries** (init, add, commit, branch, checkout, merge); content-addressed storage via hashing, custom linked-list and key-value structures, manual memory management; built with Make, documented with Doxygen.

LEADERSHIP & ACTIVITIES

Sono Barco CentraleSupélec | *Student Association* — Paris, France
Treasurer & Event Technician — 2025 – 2026

- Managed **€120k annual revenue** and tax filings; led event setups with **€300k+ of equipment** for up to **4,500 attendees**.

SKILLS & LANGUAGES

Languages: English (C1), French (native), Spanish (B1)
AI & Data Science: Deep learning (Transformers, Neural ODEs), physics-informed ML, time-series modelling, PyTorch, Scikit-learn
Programming: Python (NumPy, pandas), C, C++ (basics), SQL, Bash, TypeScript (NestJS, React), Java
Systems & Tools: Linux, Git, Docker, MPI, OpenMP, SLURM, Spark, AWS
Client & Business: Stakeholder requirements gathering, contract negotiation, budgeting & accounting
Interests: Sound engineering & DJing, bass guitar, aeronautics (French Aviation Certificate, BIA), cooking